Abdülmuttalib GÜLER G181210011 2. öğretim A grubu

1. Soru

1. Tur

$n=4$ iken

for $j=1$
$j=2$
$j=3$
$j=4$

$n=2$ olacak

2. Tur

$n=2$ iken

for $j=1$
$j=2$
end for

$n=1$ olacak

3. Tur

$n=1$ iken

for $j=1$
end for

$n=1/2$ end while

n $\quad$ $n/2$ $\quad$ $n/4$

$n=4$ için 3 kez olacak

$n=2^k$ olursa

$\boxed{k = \log_2 n}$

$$\sum_{j=0}^{k} \frac{n}{2^j} = n \underbrace{\sum_{j=0}^{k} \frac{1}{2^j}} = n.\left(2-\frac{1}{2^k}\right) = n.\left(\frac{1}{2^{\log_2 n}}\right)$$

$$\boxed{\sum_{j=0}^{k}\left(\frac{1}{2}\right)^j = \frac{\left(\frac{1}{2}\right)^{k+1}-1}{\frac{1}{2}-1}}$$

Abdülmuttalib GÜLER G181210011 2. Öğretim A grubu

2. Soru

$$2^{2^{n+1}} > (x+1)! > e^n > 2^n > n\log n > n > 4^{\log n} > (\log n)^{\log n}$$

$$> \log n^{*} \log n > 2^{\log n} > 2^{\sqrt{2\log n}} > \sqrt{\log n}$$

Abdülmuttalib GÜLER G181210011

2. Öğretim A grubu

3. Soru

a) $T(n) = 4T(n/2) + \log n$

$a=4,\ b=2,\ x=0,\ y=1$

$4 > 2^0 \Rightarrow T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_2 4}) = \boxed{\Theta(n^2)}$

b) $T(n) = 4T(\frac{n}{2}) + n^2$

$a=4,\ b=2,\ x=2,\ y=0,\ b^x = 2^2$

$4 = 2^2 \Rightarrow y > 1 \Rightarrow T(n) = \Theta(n^{\log_b a} . \log^{y+1} n)$

$= \Theta(n^{\log_2 4} . \log^1 n) = \boxed{\Theta(n^2 . \log n)}$

c) $T(n) = 3T(\frac{n}{2}) + n^2$

$a=3,\ b=2,\ x=2,\ y=0,\ b^x = 2^2 = 4$

$3 < 2^2 \Rightarrow y \geq 0 \Rightarrow T(n) = \Theta(n^x . \log^y n)$

$= \Theta(n^2 . \log^0 n) = \boxed{\Theta(n^2)}$

d) $T(n) = 0.5\,T(\frac{n}{2}) + \frac{1}{n}$

$a=0{,}5,\ b=2,\ x=-1,\ y=0$

$a \geq 1$ olmalıdır, $x \geq 0$ olmalıdır. Bu sebeple bu soruda master teoremi uygulanamaz.